

Michal Zajac

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## Word Embedding and Visualization

I trained a Word2Vec model on the Text8 dataset, which is a subset of Wikipedia text that has enough variety to build a decent vocabulary and capture relationships among English words. Additionally, I trained it using a CBOW neural network from gensim, which is different from the skip-gram approach [1, 2]. The training loss decreased steadily over ten epochs, suggesting that the model was learning meaningful patterns. Then I explored some word similarities, analogies, and specific vocabulary checks. For instance, “computer” was closely related to “computers,” “computing,” and “hardware,” while “man” and “woman” had a high similarity of about 0.696. The classic analogy “king - man + woman” yielded “queen,” showing that the model recognizes gender relationships. I tested “plumber,” but to my surprise, it was not found in the vocabulary.

When examining historical figures, “washington” and “lincoln” ranked highly in similarity, probably because they are both American presidents, and “newton” and “einstein” were also close, just like “darwin” and “tesla.” I suspect this is because they all belong to the domain of science, where their names appear in similar contexts. More interestingly, looking at “responsibility” gave me neighbors like “obligation,” “authority,” and “privilege,” which seemed appropriate. On the political side, I checked “communism” and found it surprisingly similar to “capitalism,” although in practice these ideologies are generally seen as opposites. Perhaps this happens because texts often mention them in the same discussions, so the embedding learns that they share topical proximity rather than ideological equivalence.

For visualization, I applied PCA and t-SNE to see how words cluster in two-dimensional space. Both methods grouped similar words together, but t-SNE tends to form tighter, sometimes more separated clusters because it focuses on preserving local neighborhoods. PCA, by contrast, is linear and spreads out the variance along principal components, so it can be more global and sometimes less visually distinct for smaller clusters. Overall, both methods showed that words with related meanings tend to lie close together in the embedding space, demonstrating that the model captures semantic relationships in a tangible way.

## References

- [1] <https://medium.com/data-science/nlp-101-word2vec-skip-gram-and-cbow-93512ee24314>
- [2] <https://www.baeldung.com/cs/word-embeddings-cbow-vs-skip-gram>

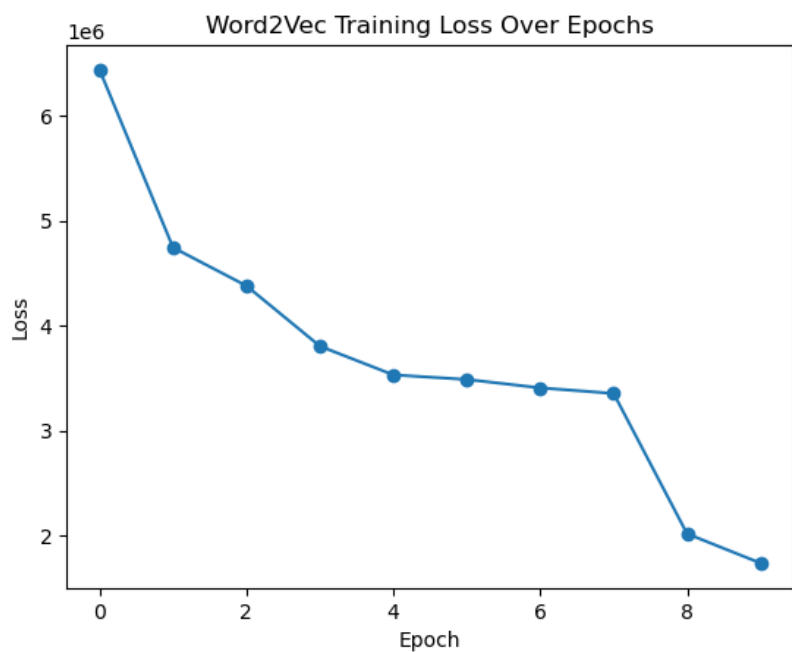


Figure 1: Training Loss over Epochs

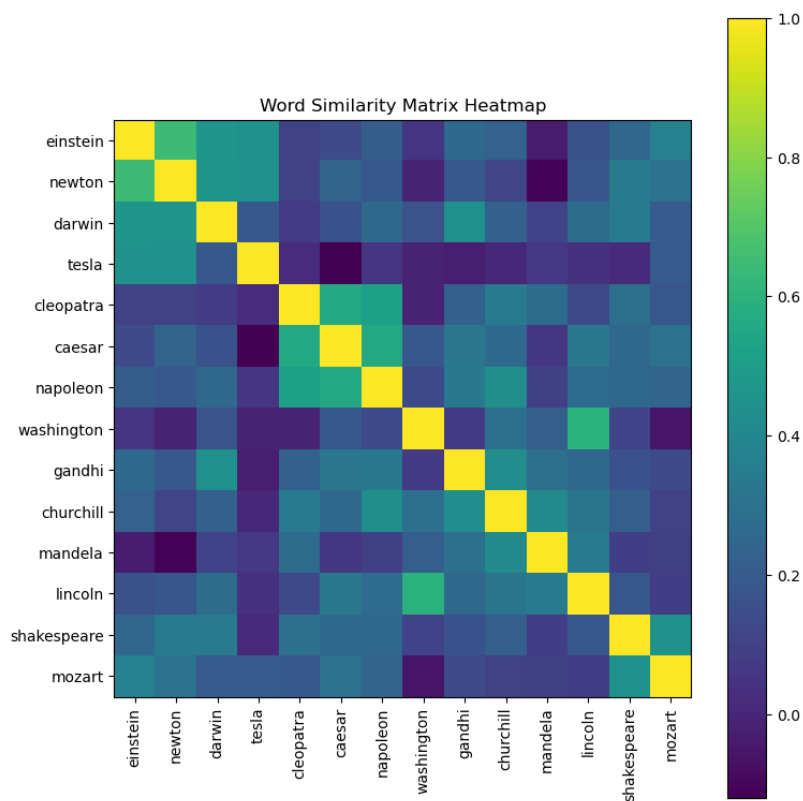


Figure 2: Historical Figures Similarity Matrix

Word Embeddings (t-SNE, 2D) (TSNE, 2D)

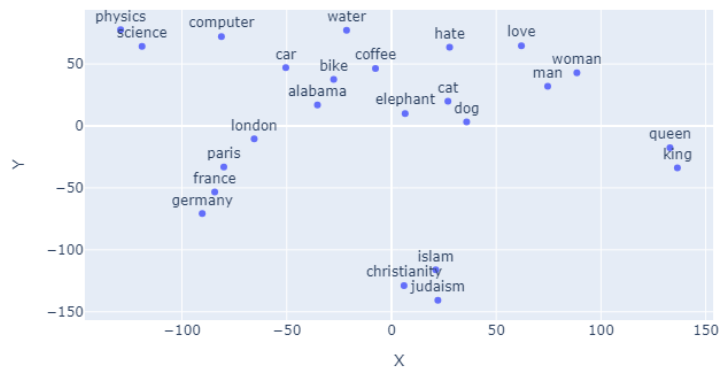


Figure 3: t-SNE Visualization

Word Embeddings (PCA, 2D) (PCA, 2D)

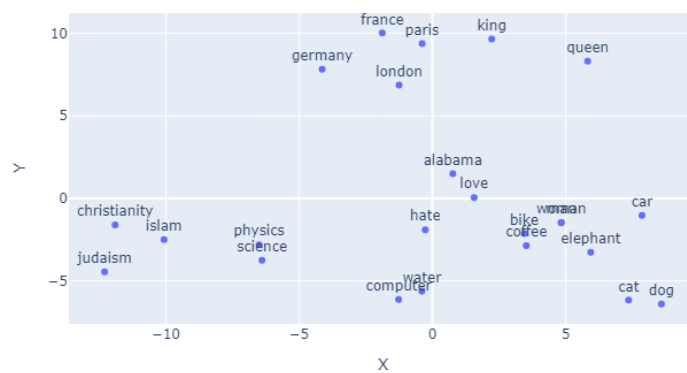


Figure 4: PCA Visualization